

MS-DOS (Microsoft Disk Operating System)

Introduction.

DOS was the main OS for Intel-based PCs from 1980 – 1995. It is supplied by **Microsoft Corporation** for use on IBM PC compatible computers or 'Clones'.

DOS is a *single-user, single-tasking* OS that is disk-based & controls your computer & the devices connected to it.

It provides the most basic instructions the computer needs to operate. It tells the computer how to process information, how to handle Input & output, and how to manage files and directories by use of commands.

- ✓ DOS is a 16-bit OS (written for 16-bit Microprocessors running 16-bit applications).
- ✓ DOS provides a Command-based interface to the user enabling him/her to execute programs. It controls the operations of your computer when you use any programs. E.g., when you create any files using DOS applications, the files are managed using MS-DOS.

DOS is normally supplied on a floppy disk or pre-installed on the Microcomputer.

DOS is used for starting the Microcomputer & managing the computer resources, which include:

- ❖ The I/O devices, e.g. Monitor, Keyboard, Printer, Floppy disk drives, Hard disk drives, etc.
- ❖ Files.
- ❖ Memory.

Memory Requirements.

The older versions of DOS required very little memory. DOS 3.0 required about 30KB of the DOS Conventional memory & less than 1MB of Hard disk space.

DOS 6.0 & above requires at least 5MB of Hard disk space & 50KB of the Conventional RAM.

BASICS OF DOS.

DOS STRUCTURE.

DOS consists of 4 basic modules: -

(i). The Boot Record.

This record begins on *Track 0, Sector 1, Side 1* on every disk formatted using the DOS **FORMAT** command. On a fixed disk, the Boot record is on Sector 1 of the DOS partition. The Boot record identifies the disk & contains the initial boot program for the disk.

(ii). The BIOS.

The **Basic Input Output System** (BIOS) is located in ROM. It is a low-level interface to the machine & is responsible for hiding the vagaries of the hardware from all other software.

The BIOS provides a series of functions that programmers can use to perform operations without having to concern themselves with the details of the underlying hardware.

(iii). The DOS Programs.

DOS is implemented by 2 programs: -

- (a). The **I/O System** - an interface module loaded from disk that augments (enhances) the ROM BIOS functions & usually contains a set of standard device drivers.

Input/Output activities - These are the *Read* or *Write* actions that a computer performs.

A computer performs a "**Read**" when you type information on your Keyboard or you choose items using a Mouse. When trying to open a file, the computer reads the disk on which the file is located to find and open it.

A computer performs a “**Write**” whenever it stores information on a disk, displays information on a Screen or sends information to a Printer.

I/O addresses - locations within the I/O address space on your computer, used by a device, such as a printer.

- (b). The **Disk Operating System (DOS)** itself. It is a high-level interface for all programs that run on the computer, whether or not they make use of the disk.

(iv). The Command Processor.

It is the normal interface to DOS for people working with the system. It generates the **Command Prompt** (C:\>), accepts commands, and executes programs requested by users of the system.

STARTING DOS (BOOTING).

Bootting is a term used to describe the startup procedure in Microcomputers.

There are 2 ways of booting a computer loaded with DOS.

- (a). From a DOS diskette.
- (b). From the Hard disk.

Starting the computer from the Hard disk.

Computers can be booted up from the hard disk if DOS has already been installed on it.

- ♣ Simply switch on the computer to start it up after making sure there is no diskette in drive A.

Note. Today, computers start or boot from Windows loaded onto your hard disk.

To start DOS, use the **MS-DOS Prompt**.

Starting the computer from a DOS Diskette.

When a computer has no hard disk, you must load DOS on diskettes.

A **DOS diskette** contains the DOS commands needed to start & use the computer.

- ♣ Insert the DOS diskette in drive A: (default drive) & then turn on the power switches on your computer.

Types of Booting.

There are 2 types of booting, namely;

- (i). **Cold booting** - Turn on all the computer power switches.
- (ii). **Warm booting** - Press **CTRL+ALT+DEL** to restart the computer.

Note. Pressing CTRL+ALT+DEL can also be used to quit an application that has stopped responding to the system.

- 1). When Power is turned on, the computer first executes a small program, **POST (Power-On-Self-Test)** that is permanently stored in ROM.

POST instructs the computer to run diagnostic tests like checking the *computer memory (RAM)* to make sure it is operating correctly; check the *CMOS (BIOS)*, check the *Hard disk Controller, Floppy disk drive Controller & the Keyboard*. If a problem is found, it displays an error message on the screen indicating where the problem is located.

- 2). If no problem is found, the computer starts looking for a “**Bootable disk**” (a disk that has the DOS “**bootstrap**” program on it). It starts with default drive (*drive A:*), followed by drive C.
- 3). If the bootable disk is found, a program in ROM executes 2 hidden Operating system programs; **IO.SYS** & **MSDOS.SYS** found in DOS.
IO.SYS is a system boot file, while MSDOS.SYS gives the booting information, i.e. path, Multi-boot, etc.
- 4). **COMMAND.COM** is then loaded into RAM.

Command.com allows the user to execute any Internal DOS command.

- 5). The computer then searches for and reads the **CONFIG.SYS** file.
Config.sys provides information about a computer configuration to the CPU. E.g., what devices the computer system has & how many files can be opened at once.
- 6). The Command.com then looks for an **AUTOEXEC.BAT** file. If an Autoexec.Bat file does not exist, Command.com will ask the user for the correct date and time.
- 7). After the date & time prompts have appeared, a **DOS Prompt** (C:\> or A:\>) appears on the Monitor. Seeing a C:\> or an A:\> means that;
 - DOS has successfully booted.
 - The default drive is the one with the letter shown.

DISK DRIVES.

A **Disk drive** is a unit that houses a disk.

Files can be stored either on the *Fixed disk drive* (usually **Hard disk** or **Drive C:**) or on *Floppy disk drive* (**Drive A:**).

In order to store or retrieve information from a disk, the appropriate drive must be specified; otherwise, DOS will assume the current drive.

Default drive.

This is the drive the computer checks first or returns to if no other specific instruction is given for it to use another drive.

Most microcomputers use drive A as their default drive when starting up.

When a microcomputer is switched on, it normally looks for DOS in drive A first. If it doesn't find DOS in drive A, then it checks the hard disk.

To change the default drive after starting the computer, type the *drive specifier* and a *colon*.

E.g. **A:** {press ENTER} will change to A:\>.

DOS Prompt.

The symbol C> or C:\> with a blinking cursor after it is called the “**DOS Prompt**”.

When you see the DOS Prompt, it means that DOS is waiting for you to enter a DOS command or the name of the program to run.

Descriptive Questions.

1. (a). What do the initials MS-DOS stand for?
(b). What type of operating system is DOS?
(c). What is the function of DOS?
(d). How many users can DOS handle simultaneously?
2. List the four basic modules of DOS?
3. Where is the boot record loaded on a hard disk?
4. (a). What does BIOS stand for and where is it located?
(b). What is the function of the BIOS?
5. (a). What generates the Command Prompt in DOS?
(b). What is the function of the Command Prompt?
6. Briefly explain the sequence of operation during boot-up.
7. What is the default drive in most computers and what does the word ‘*default*’ mean in this context?

FILES & FILE STRUCTURES.

What is a File?

A *File* is a collection of information initially created in memory and then stored with a filename on a secondary storage device for preservation & future reference.

Creating a file means grouping related information so that it can be easily accessed.

Types of Files

There are three different types of files:

System files

The operating system needs system files to run.

Examples of system files:

a). AUTOEXEC.BAT

This is a *batch file* (i.e., a file which contains a list of commands), that is run whenever you start your computer. However, it is not needed by the computer for it to be able to run.

b). COMMAND.COM

This file is very important in the startup of your computer. It contains the user interface for Microsoft DOS (i.e., the DOS prompt, and all common commands).

c). WIN.COM

This file is essential to Windows systems as it is used to start up the GUI (Graphical User Interface) for Windows. Without it Windows cannot start, even if the file was just accidentally renamed or moved.

d). CONFIG.SYS

Config.sys is not needed to run the computer, but it can be used to tell Windows or DOS specific information about the hardware that you are using, or to create a simple startup menu.

e). IO.SYS

This is one of the most important system files, as it tells your computer how to communicate with its hardware on the lowest level. Without it, nothing will happen on your computer, and Windows will not start.

f). MSDOS.SYS

MS-DOS.SYS is a basic system file for DOS. It is similar to IO.SYS in its functions, and still needs to be included for compatibility.

g). USER.DAT

USER.DAT is a data file that can be found in the Windows directory of your computer. It is needed together with SYSTEM.DAT for the Windows registry (a large database containing all the information about your computer's hardware & software).

Application (program) files

They are used to run application programs & are installed onto your computer's hard disk.

Document (user) files

These are files that contain user information and are created using application programs.

File Management.

This is the organization & maintenance of files in a computer storage media.

File management enables the user to arrange his/her files into groups according to types; thus speeding up storage & retrieval of information.

Naming of Files.

Filename.

A **Filename** is a reference name given to a file when it is saved.

A file must be given a unique name in order to help the user differentiate it from other files (especially if there are many files stored on one disk drive) & also it helps one to remember what is inside the file.

A filename consists of a '*file-name*' and an '*extension*'.

The file-name must not exceed 8 characters & can contain any alphanumeric characters, A-Z, 0-9 except the following: -

Full stop (.), Square Brackets ([or]), double quote ("), colon (:), semicolon (;), slash (/), backslash (\), less than or greater than sign (< or >), question mark (?), plus sign (+), equal sign (=), asterisk (*), or **blank space**.

You can also mix numbers, letters and symbols when entering a filename.

File extension.

An extension enables the user to group files into categories or file types by giving more information about what is in the file, such as Text (.TXT) files or Command (.COM) files.

Rules that apply to filename extensions.

- (i). Use a **period/dot** (.) followed by a maximum of 3 characters for the extension.
- (ii). Don't use any spaces between the filename and the extension.

Note. DOS reserves certain filenames for specific devices that the computer uses, and therefore, they cannot be used when naming files.

They include; AUX, CLOCK, COM, CON, KEYBD, \$, APT, PRN, etc. These names can be used as extensions but not as filenames.

DOS DIRECTORIES & DIRECTORY STRUCTURES.

What is a Directory?

A *directory* is a unit that groups related files for storage in a disk.

Directories are stored on the disk with files in them.

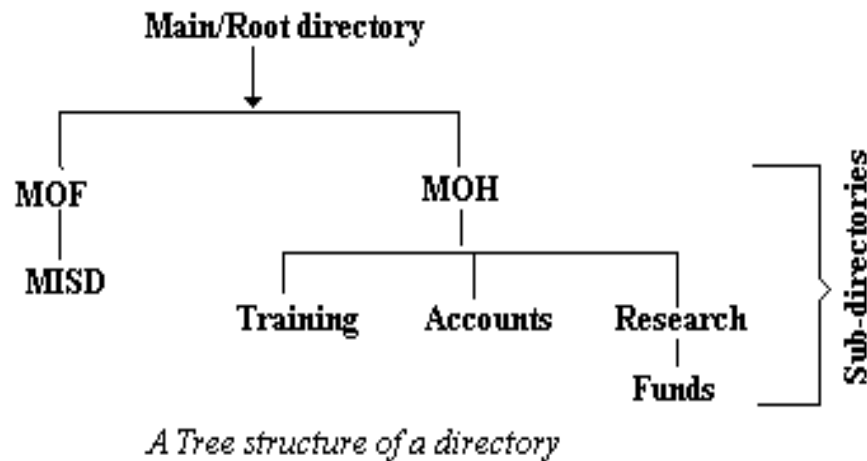
Directory Structure.

This is a method of organizing files so that one can efficiently store and retrieve data.

DOS uses a **Tree (hierarchical)** structuring system that allows the user to easily store and manipulate a large number of files efficiently.

Every disk formatted by DOS contains a directory called the **Root directory**. The Root directory is the main (top-level) directory & is used to hold a list of all the files stored on a disk.

You can create separate directories called **Sub-directories** under the main/root directory or within other directories & use them to hold files that you want to keep separate from those in the root directory.

**Root (Main) directory.**

This is the directory in a disk drive from which all other directories branch.

It is indicated by a **Backslash** (\). E.g., C:\ or A:\

Subdirectory.

Any directory other than the Root directory.

Uses of Subdirectories.

- (i). Can be used to store files that are not required currently until they are needed again.
- (ii). Can be used to store the files of one program separately from the files of another program.
- (iii). It allows each user to keep his/her files separately, even if all the users are using the same system.

From the above graph;

The Root directory has 2 subdirectories; **MOF** and **MOH**.

The subdirectory **MOF** has a subdirectory **MISD**, while the subdirectory **MOH** has 3 sub-directories; **Training**, **Accounts** & **Research**. The subdirectory **Research** has the subdirectory **Funds**.

Current (Default or Active) directory.

This is the directory that you are currently using or working in.

When you turn on the computer, the Root directory is the current directory until you change to another directory.

DOS will access the current directory when you specify a file without specifying a directory.

The TREE command.

The **TREE** command displays graphically the structure of a directory.

TREE [drive:] [path][/F]

Drive – specifies the disk for which you want to display the directory structure.

Path – specifies the directory for which you want to display the structure.

/F – used to display the names of all the files in each directory.

PATH & PATHNAMES.

A **Path** is a list of directory names separated by backslashes that specify the location of a file within the **Directory Tree** (A graphical display of the directory structure in a disk).

It is the route MS-DOS must travel, starting at the root directory to get the files in another directory.

Before DOS can find a file in a directory (other than the current directory), you must tell it where to locate the file by specifying the **Pathname**.

E.g., a pathname can be a drive specifier, a series of directory names & a file specification. Each of the elements is separated by a *Backslash*.

A:\directory1\directory2\filename.extension

A – The drive.

Directory 1, 2 – name of your directory. You can enter as many directory names as necessary.

Filename – name of the file you want to locate.

Extension – type of the file.

E.g., to specify the path of a file named README.WRI located in the WINDOWS directory, you would type,

C:\>**Windows\Readme.Wri** ↵

Filespec.

This refers to the full file specification when a drive letter is included. A full file specification usually consists of the drive letter, a colon & the filename.

The drive specifier must be placed before the filename. It tells DOS where the file is located.

E.g. A:\MOF\Sample.Txt

Descriptive Questions.

1. What is a Computer file?
2. Differentiate between a file and a directory.
3. (a). A file name consists of three parts. Name them.
(b). What characters can be used when naming a file?
(c). List at least three reserved words in DOS that should never be used as a filename.
4. Define the word 'Filespec' and give an example of a filespec.
5. With the help of a diagram, describe the directory structure of DOS.
6. What are the uses of sub-directories?
7. What is the purpose of the PATH command in DOS?
8. Name the command that shows the structure of a directory like the branches of a tree.

MS-DOS COMMANDS.

What is a Command?

A **Command** is an instruction issued by the user to the computer telling DOS what to do.

DOS commands are provided in DOS diskettes bought with each microcomputer.

Other than the DOS commands, every application program may have its own sets of commands to perform specific tasks.

Internal commands of DOS.

These are the simplest & the most commonly used commands. They are stored in the Main memory for fast access.

The Internal DOS commands are found in **COMMAND.COM** file. The commands cannot be seen when you list the directory (using **Dir**), because they are part of the **Command.com** file. They are executed via the Command.Com file, which is loaded automatically into RAM when DOS is booted.

COMMAND.COM (the most important DOS command) is used to start up the machine. It contains Internal commands such as "**Dir**" and "**Prompt**". It is responsible for accepting user commands & interpreting them for DOS.

Examples;

DATE TIME CD / CHDIR MD / MKDIR RD / RMDIR
 TYPE - used to have file contents displayed or printed.
 DIR - used to obtain a directory listing (to view directories).
 COPY - used to duplicate one or more files and save it.
 CLS - used to clear the screen. On typing this command, everything on the screen gets cleared off & the DOS Prompt appears on the top left corner of the screen.
 DEL - used to delete one or more files from disk.
 VER - used to find the version number of DOS that is in use.
 REN / RENAME - used to rename files.

DATE & TIME commands.

DOS uses the computer clock to keep data and time information.

The date information is given as **mm/dd/yy** (where **mm**=month; **dd**=day; **yy**=year).

The time information is given as **hh:mm:ss** (where **hh**=hours; **mm**=minutes; **ss**=seconds).

(a). To display or set the computer's date, type:

DATE {without parameters} ↵ - to display the current date setting & prompts for a new date. To keep the displayed date, press ENTER.

◆ **To verify the date.**

E.g., **To change the date to 11/28/2003 or 1/1/99**, type:

DATE ↵ then type:

11-28-2003 or 01-01-99

(b). To display or set the computer's time, type:

TIME {without parameters} ↵ - to display the current time setting & prompts for a new time. To keep the current time, press ENTER.

◆ **To verify the time.**

E.g., **To change the time in the system to midnight or 8 pm or 11 minutes past 10 am**, type:

TIME ↵, then type:

12:00a (for midnight) or **08:00p** or **10:11:20a**↵

External commands of DOS.

The External DOS commands are stored within the **DOS directory** on a secondary storage device, such as hard disk, from where they can be called when required. This means that, they cannot be accessed directly when the computer boots up.

Examples.

DISKCOPY.COM – Copies the contents of one floppy diskette to another.
 FORMAT.COM - Formats diskettes for use with MS-DOS.
 BACKUP.COM – Copies files from a fixed disk to a diskette.
 DISKCOMP – Compares the contents of two floppy diskettes.
 PRINT.EXE

When you enter a command, DOS checks its library to see if the command exists. The sequence of search is:

- Main memory - Internal commands.

- Secondary storage - External commands.

If the command is found then it will be executed. Otherwise, an error message will be displayed, such as “*Bad command or Filename*”

DOS Command Line.

The DOS Command Line is where users “type” commands. The command line is indicated by a DOS Prompt, i.e.,

A:\> or **C:\>** and a **blinking cursor**.

The letters ‘A’, ‘B’, and ‘C’ refer to disk drives.

Rules of typing MS-DOS commands.

- ☞ Commands can be typed either in lowercase or uppercase.
- ☞ Incorrect spelling or spacing will not be accepted by the DOS language.
- ☞ To correct typing mistakes, press **Backspace** key and retype the command.

Entering of commands is done in 2 steps:

- (i). Type the command & if necessary edit/change it using the command line editing keys, e.g. BACKSPACE for erasing typed input.

- (ii). Press the ENTER key.

When you press the Enter key,

- The line you typed is handed over to the DOS Command processor (**Command.com**) for processing.
- The line is then saved in the command line TEMPLATE. The TEMPLATE can only save one line, so each new line you type replaces the last line saved.

The TEMPLATE feature saves DOS users’ time in typing immediate commands that have already been typed.

Function Editing Keys;

F1 - Copies one character at a time from the template.

F3 - Copies the entire template to the screen.

F5 - Moves the cursor down one line to allow you to edit or retype the currently displayed line.

GETTING HELP.

To get assistance on how to use any of the DOS commands;

- ❖ At the Command Prompt, type the **name of the command** followed by a **space**, the **/?** switch, then press **ENTER**.

E.g., **FORMAT/?** ↵ will display the syntax of the **Format** command including all the switches and their usage.

Using Wildcards.

A Wildcard acts as a substitute for a filename or extension.

There are 2 wildcards: -

Asterisk (*) – represents a whole word or a group of characters.

Question mark (?) – represents a single character.

E.g., To list all files with .COM extension in the DOS directory, type:

C:\DOS>DIR *.COM ↵

Descriptive Questions.

1. Why are a few commands loaded into computer-memory?
2. (a). What are the two classes of DOS commands?

- (b). What are the differences between Internal and External MSDOS commands?
 (c). List 3 examples of Internal and 3 External commands.
 (d). Name the command used to start up the computer.
- Suppose you change the year of a microcomputer clock using the DATE command from 2001 to 2003 and then switch it off. What will be the date indicated by the microcomputer when it is switched on again after a few minutes?
 - What is the main difference between 'deleting' and 'backspacing'?
 - What is a Template? How do you recall the last command entered via the keyboard?

FILE COMMANDS.

Dir	Type.	More.	Copy.
Move.	Del or Erase.	Undelete.	Print.
Rename or REN.			

1). LISTING (DISPLAYING) FILES IN A DIRECTORY OR DISK.

- (a). To list all the files contained in the current directory, type:

DIR ↵

- (b). To list the files in the root directory in your floppy disk (drive 'A'):

At A:\>, type: **DIR ↵**

-Or-

At C:\>, type: **DIR A:\ ↵**

- (c). To list the files in the directory MOF in drive C:

At C:\>, type: **DIR C:\MOF ↵**

The directory listing (or the DIR command) gives you 5 columns of information:

- Name of the file.
- Extension (if any).
- File size (no. of bytes).
- Date the file was created or updated.
- Time the file was created or updated.

Note. DIR has 2 options; /W and /P. These two options make file or directory listings easier to read.

/W (WIDE) - causes the directory to be displayed horizontally across the screen.

/P - pauses the directory listing once the screen is filled.

- ❖ To display the contents of drive A, pausing after every screen is full:

At A:\>, type: **DIR/P ↵**

-Or-

At C:\>, type: **DIR A:/P ↵**

- ❖ To display the contents of drive A, horizontally across the screen (or using the Wide format display):

At A:\>, type: **DIR/W ↵**

-Or-

At C:\>, type: **DIR A:/W ↵**

2). DISPLAYING TEXT FILES.

ASCII text files can be displayed using the **TYPE** command.

Syntax: **TYPE** <filename.extension> ↵

The MORE command.

The MORE command is used to pause the screen display of files or directory listing, i.e. it allows you to view a screen page at a time.

Syntax: **TYPE** <filename>|**MORE** ↵

-Or-

MORE <filename>

3). CREATING FILES USING DOS.

Files can be created directly from the DOS Command Line using the **COPY CON** (Console) command.

If you type **COPY CON** followed by a filename, all inputs received from the Console or keyboard (CON) will be copied into the designated file.

After the last entry, close the file and write it to the disk by pressing **F6** (or **CTRL-Z**) and <**ENTER**> keys.

E.g., to create the file 'SAMPLE.TXT' on a disk and enter the text 'red, blue, green', type the following:

A:\>**COPY CON Sample.Txt**{press ENTER}

Red {press ENTER}

Blue {press ENTER}

Green{press ENTER}

Press **CTRL-Z** or **F6**, then press **ENTER**

The latest DOS versions have an editor called **EDIT** that can be used to create or edit MS-DOS text files. The MS-DOS Editor is a full-screen editor, which allows you to create, edit, save and print ASCII text files.

To create a file using **EDIT**, type:

C:\>**EDIT** <filename> ↵

4). DELETING FILES.

Files that are no longer needed take up space on the hard disk or floppy disk. To delete these files, use **ERASE** or **DEL** commands.

Note. Deleting files is dangerous, because you can sometimes delete files that you still need. Make sure that you only delete files that are no longer in use.

To delete a file, you must specify the filename & its extension (if any), together with the directory in which the file is located. Otherwise, DOS will assume the current directory.

Examples:

- (a). To delete the file named MEMO1 from the current directory.

DEL MEMO1↵

- (b). To delete all the files with an extension .DBF from a directory named MOF in drive A.

DEL A:\MOF*.DBF {ENTER} -Or- change to A:\MOF>, then type: **DEL *.DBF** {press ENTER}

- (c). To delete all the files from the root directory on drive C:

DEL C:*. * { ENTER} -Or- change to C:\>, then type: **DEL *. *** {press ENTER}

- (d). To delete a file named SAMPLE1.TXT on drive A:

A:\>**DEL SAMPLE1.TXT** {press ENTER}

- (e). To delete all the files in C:\TEMP directory;

C:\TEMP>**DEL *. *** ↵ -Or- C:\TEMP>**DEL** ↵

Tip. /P –prompts for confirmation before deleting each file.

Note. Subdirectories cannot be deleted by the DEL command. To delete all the files & Subdirectories within a directory use the **DELTREE** command.

UNDELETE command.

The MS-DOS **DELETE** command does not actually erase the file from the hard drive. As long as the Sectors containing the file pieces are not actually overwritten, the file is retrievable.

In MS-DOS 5.0 or 6.22, you can recover files deleted accidentally using the **UNDELETE** command. However, this command will only work if the space occupied by the file has not been used by another file. Therefore, if you delete a file accidentally, make sure that you undelete it before performing any other tasks on that particular disk.

To undelete a file,

Run the **UNDELETE** command (without any switches) as soon as possible after you have deleted the file. If the sentry directory is present, undelete shows you the files available for recovery.

To undelete the file called SAMPLE1.TXT from drive A, type:

```
A:\>UNDELETE Sample1.Txt↵
```

You will be prompted to provide the first letter of the filename again.

5). RENAMING FILES.

To change the name of a computer file, use the **REN** or **RENAME** command.

E.g., to change the name of a file named SAMPLE.TXT in drive A to LAB.DAT, type:

```
A:\>RENAME Sample1.Txt Lab.Dat↵
```

6). PRINTING FILES.

A file may be sent to the printer by specifying the device name of the printer in the *Copy* command or by using the **PRINT** command. The printer device name is usually designated **PRN**.

To print a file named SAMPLE.TXT, type:

```
A:\>COPY Sample1.Txt PRN↵
```

-Or-

```
A:\>PRINT Sample.Txt↵
```

7). COPYING FILES.

The **COPY** command copies one or more files to a different directory on the same disk or to another disk.

To copy a file, specify the Source directory, the filename, its extension, & the Destination directory. In case the source or destination is not specified, MS-DOS assumes the current drive or directory.

Source directory – the directory that contains the file(s) you intend to copy or move.

Destination directory – the directory to which you intend to copy or move one or more files.

DOS automatically creates the destination file during the copying process.

Examples:

- (i). To copy the file named MEMO1 on drive C to a directory named MOF on the same drive.

COPY C:\Memo1 MOF ↵

- (ii). To copy a file named MEETING from a subdirectory LETTERS within the subdirectory MOF on drive A to a subdirectory named MOF on drive C:

COPY A:\MOF\Letters\Meeting C:\MOF ↵

- (iii). To copy the file named MEMO2 from the current directory to the root directory on drive A:

COPY Memo2 A:\ ↵

- (iv). To Copy all the files from the root directory on drive A to the current drive, current directory.

COPY A:*.* ↵

- (v). To copy all files with an extension .DBF from a directory named MOF in drive A to a directory named MOF in drive C.

COPY A:\MOF*.DBF C:\MOF ↵

-Or-

Change to A:\MOF\>, then type: **COPY *.DBF C:\MOF ↵**

- (vi). To copy the CONFIG.SYS file from drive C to drive A:

C:\>COPY Config.Sys A: ↵

Tips.

/V –verifies that the new files are written correctly.

/-Y –causes prompting to confirm you want to overwrite an existing destination file.

Wildcards can be used to copy a group of files. For instance, to copy all Command files (files with extension **.COM**) from the DOS directory in drive C to drive A:

C:\DOS>COPY *.COM A: ↵

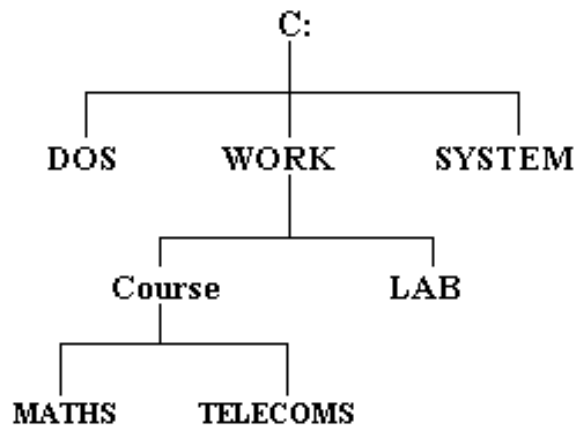
Note. To copy all the files in a directory, including the files in the subdirectories of that directory and also hidden files, use the external **XCOPY** command.

At C:\>, type: **XCOPY C: A: ↵**

Descriptive Questions.

1. State four examples of File commands.
2. What is the function of the MORE command in DOS?
3. What facility does DOS provide to create ASCII text files?
4. Give the command used to:
 - (a). Display the name of the current directory or change from one directory to another.
 - (b). Copy entire directories including all the subdirectories.
 - (c). View the contents of the directory.
5. By giving examples, explain the functions of the '*' and '?' characters as used with DOS commands.

6. Consider the DOS Hierarchical directory structure shown below.



Assume that all subdirectories are empty initially. Unless changed otherwise, for each action below, start from the root directory and write down the commands required to perform the following:

- (a). Create two subdirectories under SYSTEM called LOTUS and WP.
- (b). Create a file called Maths1.txt under MATHS subdirectory with the following contents.
1+1=2
3+3=6
- (c). While in the MATHS subdirectory, copy Maths1.txt to Maths2.txt.
- (d). Remove MATHS subdirectory.
- (e). Remove LAB subdirectory.

DIRECTORY COMMANDS.

Dir CD / CHDIR RD / RMDIR MD / MKDIR

1). CREATING (MAKING) A SUBDIRECTORY.

MS-DOS creates the root directory any time it formats a disk and then specifies the current directory on your screen.

MS-DOS will indicate the root directory in a given drive by the **drive letter** followed by a **colon &** then a **backslash**. E.g., the root directory on drive A will be **A:**, while the root directory on drive C will be **C:**

To create a directory, use the **MD** command or its longer form **MKDIR**.

- (a). To create the subdirectory MOF under your root directory in drive C, and then change to the subdirectory, type:

C:\>MD MOF ↵

C:\>CD MOF ↵ (DOS will change the prompt to read C:\MOF>)

- (b). To create the subdirectory MISD under the subdirectory MOF;

Change to the subdirectory MOF, then type: **C:\>MD MISD** ↵

2). CHANGING SUBDIRECTORIES.

To work in another directory, use the **Change Directory (CD)** or **CHDIR** command.

- (a). To change from the root directory on drive C to the subdirectory MOF;

CD MOF ↵

DOS will change the prompt to read C:\MOF>

- (b). To further change to the subdirectory MISD, which is a subdirectory of MOF;

While in C:\MOF>, type: **CD MISD** ↵

(The prompt will now change to C:\MOF\MISD>)

- (c). To move to a subdirectory called PETER which is under the directory USER; type:
CD\USER\PETER {press ENTER}
- (d). To move one level of directories backwards, use
CD.. (CD followed by two periods) or **CHDIR..**
- (e). To change to the root directory (C:\>), use
CD (CD followed by a backslash).

3). COPYING FILES BETWEEN DIRECTORIES.

When subdirectories are created, they are empty. You can copy or create files in them. To copy files to the directory, use the **COPY** command.

When copying files between subdirectories, the full path of where the file is coming from (source) and where it is to be copied (destination) must be included.

To copy SAMPLE1.TXT file from PETER subdirectory in drive A to USER subdirectory in the same drive;

A:\PETER>**COPY Sample1.TXT USER**{press ENTER}

4). REMOVING DIRECTORIES.

Like files, directories can also be deleted.

A directory can be removed or deleted using the **Remove Directory (RD)** or **RMDIR** command.

Rules (precautions) to take when deleting or removing a directory.

- ❖ You can delete a directory only if it is empty, i.e. it has no files or subdirectories in it.
- ❖ You cannot delete/remove the current directory or the root directory. You should be in the subdirectory one level higher than the directory to be removed.

E.g., before you can remove a subdirectory called WINNIE in drive A which contains files, you must delete its contents first.

To remove the subdirectory WINNIE,

In A:\WINNIE>, type: **DEL *.***↵ (to remove all files).

A:\WINNIE>**CD..**↵ (to go one level up).

A:\>**RD WINNIE**↵ (to remove WINNIE directory).

Assuming that you have the following directories C:\MOF\MISD>. To delete the subdirectory MOF, you must first remove the subdirectory MISD.

To remove the subdirectory MISD; at C:\MOF\MISD>, type:

RD\MOF\MISD {press ENTER}

-Or-

Change to C:\MOF>, then type: **RD MISD** {press ENTER}

You can then remove the directory MOF by typing: **RD MOF**↵ at the root directory.

Exercise (a).

1. What is the function of the following MS-DOS commands?
 - (i). CD/CHDIR
 - (ii). DIR.
 - (iii). TREE.
 - (iv). REN/RENAME.
 - (v). RD.
 - (vi). DELTREE.
2. What two rules must be followed before deleting a directory?

Exercise (b).

1. For each of the following MSDOS file management tasks, state one specific command that can perform the indicated task.
 - (i). Deleting files.
 - (ii). Changing directories.
 - (iii). Viewing directories.
 - (iv). Creating directories.
 - (v). Removing directories.
 - (vi). Renaming files.

WORKING WITH DISKS.

Types of Diskettes.

The 2 types of diskettes most commonly in use are:

- 3½-inch diskettes (720KB, 1.44MB).
- 5¼-inch diskettes (360KB, 1.2MB).

Uses of Diskettes.

Diskettes are mainly used for:

- (i). Backing up files for safekeeping.
- (ii). Backing up files that are not being used currently, to leave space on the hard disk.
- (iii). Transferring files from one machine to another.

Write Permit/Protect Notch.

Diskettes are provided with a write protect notch. The notch prevents writing to the diskette.

In 5¼" diskettes, the write-protect notch is covered with a sticker. In 3½" diskettes, the write-protect tab is slid towards the front of the diskette.

Checking a Disk – CHKDSK or SCANDISK.

CHKDSK command.

CHKDSK normally checks the status of a disk, fixes errors in it & displays a status report.

The report shows which errors were found in MS-DOS 6.2 filing system, which contains the FAT and directories.

It also displays a summary information on the disk space used, disk space available & bad sectors (if any). Otherwise, it gives an error message if errors are found in the disk.

Some of the information returned by the CHKDSK command includes:

- *Disk information:* Volume name, date created, total disk space and the space used.
- *Main memory information:* total bytes and bytes available.

- (i). To check the status of drive A that contains data files & display the status report.

At C:\>, type: **CHKDSK/F A:** ↵ -Or- at A:\>, type: **CHKDSK/F** ↵

- (ii). To check the status of the current disk (hard disk) that contains data files, type:

C:\>**CHKDSK/F** ↵

The /F switch – tries to fix errors on a disk when using the CHKDSK command.

Note. Running **CHKDSK/F** when files are open can result in corruption or loss of data. Therefore, never run it from another program or from *Microsoft Windows*.

SCANDISK command.

SCANDISK is a full-featured disk analysis & repair program that checks a disk for errors & repairs (corrects or fixes) any problem if found.

If you suspect that files in your hard disk or floppy disk may be corrupted or there may be any other problems with your disk, run **Scandisk** (without parameters) to check for & repair errors on the current drive.

Scandisk checks & fixes problems in the following areas:-

◆ **FAT**

File system structure (lost clusters, cross-linked files). *Lost clusters* are the remains of temporary files that are not cleared if the user crashes out from Windows, & are normally of no use and take up disk space. Lost clusters are not linked to a filename, but are recorded in the FAT as holding data.

Cross-linked files are clusters on disk that are linked to more than one file.

◆ **Directory tree structure.**

◆ **Physical surface of the disk (bad clusters).**

◆ **MS-DOS boot sector.**

If SCANDISK finds a problem, it displays a **Problem found** dialog box that briefly explains the problem and how to fix it. For the computer to fix the problem, choose the **Fix** button.

To check & repair a drive, use the following syntax:

SCANDISK [drive:]/All/AutoFix/NoSave/NoSummary/Surface↵

/ALL – checks & repairs all local drives.

/AUTOFIX – fixes damage without prompting.

/NOSAVE – deletes lost clusters rather than saving as files.

/NOSUMMARY - prevents Scandisk from stopping at summary screens.

/SURFACE – performs a full surface scan for your drive after other checks.

/CUSTOM – configures & runs Scandisk according to SCANDISK.INI settings. **Scandisk.ini** is a disk status tool configuration file.

/MONO – configures Scandisk for use with a Monochrome display.

File System - the overall structure in which files are named, stored & organized by the OS.

E.g., MS-DOS & earlier versions of Windows use FAT16 file system. Windows 98 can use FAT16 or FAT32 file system.

File Allocation Tables (FAT) - A method used by OS to keep track of where files are stored on a hard disk.

Checking Memory Usage – MEM

The memory configuration of a computer significantly affects the system's performance. It is therefore important to know how much & what type of memory your computer has.

The total amount of memory is usually shown in the upper left-hand corner when the computer boots up.

To find out the memory configuration, use the **MEM** command, which displays information about allocated memory areas, free memory areas, the total amount of memory used & the programs currently loaded into memory in your system.

/CLASSIFY or /C – classifies programs by memory usage. Lists the size of programs, provides a summary of memory in use, and lists the largest memory block available.

/DEBUG or /D - displays status of all modules in memory, internal drivers, & other information.

/FREE or /F - displays information about the amount of free memory left in both Conventional & Upper memory.

/PAGE or /P - pauses after each screenful of information.

Partitioning of Hard Disks – FDISK.

Hard disks (not floppies) can be divided into partitions or sections.

MS-DOS supports 2 types of partitions: -

- (i). Primary DOS partitions, and
- (ii). Extended DOS partitions.

The **primary DOS partition** is the area that stores the DOS System files. It is therefore, designated as the *Active partition*.

An **Extended DOS partition** is optional, and is used to divide the hard disk into more than one section.

To configure (or create partitions in) hard disks for use with MS-DOS, use the **FDISK** program.

❖ To view your current drive status (display partition information);

At the MS-DOS Prompt, type: **FDISK/Status**↵

-Or-

Type: **FDISK**↵, and then choose **Display Partition information**.

PARTITIONING THE HARD DISK DRIVE FOR USE WITH MS-DOS.

At C:\>, type: **FDISK** ↵

The FDISK Options menu appears. From the menu, choose one of the following:

1. Create DOS Partition or Logical DOS Drive.
2. Set Active Partition.
3. Delete Partition or Logical DOS Drive.

To exit FDISK, press **ESC**.

Creating DOS Partition or Logical DOS Drive.

Enter choice No. **1**, press ↵ then choose one of the options displayed depending on what kind of partition you want to create.

To create a Primary DOS Partition;

Enter choice No. **1** from the menu that appears.

“Do you need maximum size for DOS Partition? (Y/N)

Choose “**Y**”, if you want to use the entire hard disk for MS-DOS. FDISK will then prompt you to insert a DOS diskette in Drive A: and restart your computer.

To create more than one MS-DOS partition, choose “**N**”. This allows you to create Primary MS-DOS Partition, smaller FDISK displays & then a menu.

To create an Extended DOS Partition.

This is the partition that is not used for the primary MS-DOS partition.

Enter choice No. **2** from the menu that appears. The following appears on the FDISK screen:

Partition	Status	Type	Size	Percentage
C: 1	PRI	DOS	Size of Hard disk	% of the hard disk size used

If you want the default no. of MB available for extended partition, press **ENTER**, type the size in MB or in % of the hard disk to be used, then press **ENTER**.

Example:

Partition	Status	Type	Volume Label	Mbytes	System	Usage
C: 1	A	PRI DOS	FIXED DISK	2471	FAT32	30%
2		EXT DOS		5750		70%

Total disk space is 8221Mbytes (1MB = 1,048,576 bytes).

Tips.

- ❖ FDISK & FORMAT are utilities necessary for installing a new hard disk in your computer or for starting over afresh with a clean disk.

FDISK is used first to create a partition & then FORMAT is used to make the partition available for use.

Warning: Using FDISK incorrectly can destroy all your data on your hard disk. Therefore, don't use FDISK if you're not sure of how to use it.

Preparing a Diskette – FORMAT.

In order for MS-DOS to be able to use a new disk, you must first use the **FORMAT** command to prepare the drive.

Formatting prepares a new disk for use with your computer & deletes any previous information stored on it.

When formatting, you must specify the drive; otherwise, DOS will assume the current drive.

The **FORMAT** command creates a new root directory & 2 File Allocation Tables (FATs); one for redundancy, on the disk. It can also check for bad areas on the disk.

Root directory - is the starting point from which all other directories branch. It keeps a listing of all files along with the address of the first cluster assigned to each file.

File Allocation Table (FAT) - is a table of entries describing how each cluster on disk is allocated. Each cluster has a specific entry and is marked as *available*, *reserved*, *bad*, *the last entry for a file*, or *used*.

Note. **FORMAT** creates an image of the disk. The 'image' includes copies of the FAT and all directory entries.

Warning:

- ❖ Format destroys (erases/removes) any existing files or information on the disk being formatted. Therefore, a disk should be formatted only if the information in it is no longer needed.
- ❖ **You should never format drive C: (or hard disk).**

The **FORMAT** command can be used to prepare an ordinary or a System disk. *Ordinary diskettes* are used for backing-up files & programs, whereas *System disks* are used for starting (or booting) the computer.

Preparing Ordinary Diskettes.

Syntax: **FORMAT [drive:]**

- (a). To format a diskette in drive A for backing up data using the hard disk, type:

C:\>**FORMAT A:/V** ↵

/V – is used to specify the volume label, which is used to identify the disk.

/B – allocates space on the formatted disk for System files.

/C – tests clusters that are currently marked “*bad*”.

/Q – performs a quick format.

Quick format deletes the File Allocation Table (FAT) & the Root directory of a disk, but does not scan the disk for bad areas.

Preparing System Diskettes.

Your computer needs the system files to start.

A **System disk** – a disk that contains the MS-DOS system files necessary to start MS-DOS. System disks are used for booting (starting) microcomputers that runs or uses the MS-DOS operating system.

There are 2 ways of preparing System disks:

- (i). Format a newly partitioned disk, and then copy System files to it.
- (ii). Add the Operating system files to a formatted disk.

Format a newly partitioned disk in drive A & copy system files to the formatted disk.

A System disk can be prepared by adding the /S switch that enables DOS system files to be copied to the diskette after formatting.

At C:\>, type: **FORMAT A:/S** ↵

System startup files will be automatically copied after your drive is formatted.

Adding System files to a formatted disk.

Use the **SYS** command to copy MS-DOS system files & Command Interpreter to a disk you specify, making the disk bootable.

Syntax: Drive1:\>**SYS drive2:**

Drive1 – the location of the system files.

Drive2 – the drive where the files are to be copied to.

♦ **To copy system files from drive C to drive A.**

At C:\>, type: **SYS A:** ↵

♦ **To SYS your drive C (to copy system files from A to C).**

At A:\>, type: **SYS C:** ↵

If the procedure is successful, a “**System transferred**” message appears.

The following system files are copied to your disk during the SYS procedure.

- (i). IO.SYS
- (ii). MSDOS.SYS
- (iii). COMMAND.COM

If the procedure is not successful, check to make sure you’re typing the correct drive letter for your drive.

To copy the Command.Com file to disk A, type:

COPY\COMMAND.COM A: ↵

The FORMATTING process.

There are 3 parameters (factors/considerations) that define how information is stored on diskettes:

Volume	A single diskette is considered to be 1 volume.
Files	Information is stored on diskettes in units called Files . Files occupy physical tracks on the disk.
Records	Information in files is further broken down into records, which occupy Sectors on a disk. A group of sectors is called a Cluster .

During formatting, the surface of a disk is broken into *Tracks* (80 tracks for high density) each with 512 bytes.

As the heads move toward the centre of the disk, each track is checked for integrity (reliability) and is divided into *Sectors* (8 or 9 sectors, depending on the type of disk).

Formatting creates 4 parts:

1. Boot sector, which is accessed during booting.
2. The File Allocation Table (FAT).
3. The Root directory/folder.
4. The Data space for storing data.

The outer track (**Track 0**) is used for storing Systems information such as boot-sector information & the **File Allocation Table (FAT)**.

FAT contains information on the location of the files in the disk. It also holds system information for booting up, especially in system diskettes.

UNFORMAT command.

The **UNFORMAT** command is used to restore a disk that was erased by the **FORMAT** command to its former state if the disk was not formatted using the **/U** switch to specify an “*Unconditional*” format. If the **/U** switch is used, then all the data on the disk is destroyed.

How UNFORMAT Works.

The **Format** command does not actually wipe out the disk sectors themselves; thus, the files are still present just as when a file is deleted. If you do not do anything with the disk before attempting to unformat it, you are able to retrieve all your files.

Syntax: **UNFORMAT [drive:]/L**

Drive: - specifies the drive to unformat.

/L - lists all the file & directory names found by **UNFORMAT**. If you do not specify this switch, **UNFORMAT** lists only subdirectories & files that are fragmented.

To unformat a high-density diskette (3½”) in drive A, using the hard disk.

At C:\>, type: **UNFORMAT A:/L** ↵

Naming a Disk – LABEL.

To create, change or delete the Volume Label of a disk, use the **LABEL** command.

Syntax: **LABEL [drive:] [label]**

Typing **LABEL** ↵ displays the current label in a specified disk & prompts you to change or enter a new volume label.

To add a volume label to drive A.

At C:\>, type: **LABEL A: KIONGWANI** ↵

VOL – Displays the disk volume label and Serial number, if they exist.

Syntax: **VOL [drive:]**

Defragmentation of Hard Disks – DEFRAG.

With time, as files are deleted and others are saved, information that is stored on the disk can become fragmented (disjointed or split). Fragmentation occurs when a file is written into non-contiguous sectors on the disk.

Fragmentation does not affect the validity of the information, i.e., the files are still complete when you open them, but it takes longer for your computer to read and write fragmented files than it does to read and write unfragmented files.

DEFRAG is a utility designed to address the problem of hard disk file fragmentation. It re-organizes the files on the hard drive to optimize disk performance.

The Defrag program works in one of two ways:

- ♦ **Unfragment files only** - This choice will restore your files to a contiguous state on the disk. However, it does not move the existing files to fill in gaps.
- ♦ **Full optimization** - This is a more effective method of reorganizing the files on your hard disk. It reconstructs all the pieces of fragmented files, moves all directions to the front of the disk and eliminates the gaps by moving all files to one continuous section at the beginning of the disk.

Making an exact copy – DISKCOPY.

The **DISKCOPY** command copies the entire contents of one diskette to another, including subdirectories, hidden files, system files and volume labels.

Warning: DISKCOPY overwrites the contents of the destination disk as it copies the new information onto it.

The DISKCOPY command can only be used to copy disks of the same type & capacity. If you only have one floppy drive, then the command will prompt you to swap the source and destination as required.

Syntax: **DISKCOPY [drive1:] [drive2:]**

/V – verifies that the information is copied correctly.

To make an exact copy of the contents of a disk in drive A on a system with 1 floppy drive;

At C:\>, type: **DISKCOPY A: A:/V** ↵

BACKUP command.

It is used to make duplicate copies of files from C to A.

At C:\>, type: **BACKUP C: A:** ↵

Advantages.

- (i). It locks your file.
- (ii). Allows you to continue from where you had reached if the destination disk gets full.

Syntax for reading Backups.

At C:\>, type: **RESTORE A: C:/S**

Note. When restoring, you have to follow the sequence.

ATTRIB command.

An external command used to make files hidden or read-only.

Syntax:

C:\ATTRIB +H - will make files hidden.
C:\ATTRIB -H - will make files not to be seen.

MSD command.

It shows the configuration of the computer.

At C:\>, type: **MSD** {ENTER}

DEBUG command.

It runs **Debug**; a program testing and editing tool. It is used in programming to remove errors.

A **Bug** is a computer error.

Syntax: **DEBUG** ↵

After Debug starts, type **?** to display a list of debugging commands.

To end the program, type letter **Q**, then press ENTER.

COMMON ERROR MESSAGES RELATED TO DISKS.**1. Not reading Drive A:****Abort, Retry, Fail?**

Abort – terminates the command.

Retry – Retries the command.

Fail – informs the user that something is wrong.

This message appears in case: -

- No diskette is drive A:
- The diskette is not correctly inserted.
- The diskette is inserted but the shutter door is not closed properly.

In case you have a disk in disk drive A and you receive this error message, then try reading drive B:

2. Not Ready reading Drive A:

Meaning – the diskette is not inserted properly or the door is not properly closed.

3. Error reading Drive A:

This message appears in case: -

- Drive A: is not properly connected.
- There is no power connected to the floppy disk drive, or
- The bus is not connected to the controller.

4. General failure reading Drive A:

The message appears if the diskette is faulty.

5. Insert diskette in drive A: and press any key when ready.

- This is caused by drive A: not configured in the CMOS Setup correctly. E.g., in the CMOS, the drive might have been configured as drive B:
- It can also happen when you want to boot & the current drive doesn't have System & startup files.

6. Non-System disk! Replace the disk and then press any key to continue.

Meaning – The disk in Drive A: is not bootable. Remove the diskette.

7. Invalid Drive specification.

Your CMOS has been erased. Try to get a disk with Command.Com.

8. LEDs keep on lighting.

Caused by a reversed bus or power cable connection to the disk drive.

9. Floppy disk drive or Hard disk drive controller failure.

Caused by a faulty Controller card.

10. Verify whether the door is open.

Occurs if drive A: is not configured in CMOS Setup.

11. Seek Error reading drive C:

Meaning – drive C: is not well connected.

12. Bad Command or Filename.

Meaning – you have given a bad command or the file is not in the directory.

13. Bad Command interpreter.

Your Command.Com is corrupt.

14. Packed File corrupt.

Your computer is infected by a virus.

HOW TO RUN PROGRAMS.

Program file - A file that starts an application or program. A program file has an **.EXE**, **.PIF**, **.COM** or **.BAT** filename extension.

Program Information File (PIF) - A file that provides information about how Windows should run a Non-Windows application. PIFs contain the name of the file, a startup directory, etc.

Document file - A file that is associated with an application. When you open a document file, the application starts and loads the file.

Non-Windows application - A term used to refer to an application that is designed to run with MS-DOS, but not specifically with Windows.

Example:

To start Windows from the MS-DOS Prompt:

1. Locate the **Windows** directory in your computer.
Assuming that, the Windows directory is located in drive C:, then
2. At C:\>, type: **Win** ↵
-Or-
Type: **CD Windows** ↵
At C:\WINDOWS>, type: **Win** ↵

Descriptive Questions.

1. What are diskettes used for?
2. List the two types of storing diskettes.
3. (a). What DOS command is used to prepare a diskette before using it for the first time.
(b). Describe the formatting process of a diskette.
4. Give the DOS commands that are used to:
 - (i). Examine the status of a hard disk and display a status report.
 - (ii). To make exact copies of diskettes.
5. What information does the CHKDSK command give?

Exercise.

1. Describe the steps a microcomputer goes through while booting up from the hard disk, once it is switched on.
2. (a). What is an MS-DOS Command?
(b). Give the difference between External and Internal DOS commands.
(c). List and explain 4 commands used in MS-DOS.
3. Name two DOS commands that may be used for checking disk errors.
4. Define the following terms: -
 - (i). DOS Prompt.
 - (ii). Default disk drive.
5. (a). Define Root directory.
(b). Give some two examples of root directories you know.
6. Write down the appropriate DOS commands to perform the following tasks.
 - (i). List/display all files in the current directory.
 - (ii). Copy all files from a directory named MICK1 to your floppy disk.
 - (iii). Delete all files with an extension .BAS in the current directory.
 - (iv). Display the volume Serial Number of a particular drive.
7. What information does the DIR command give?

8. Explain the work of the following commands.
 - (i). ERASE.
 - (ii). LABEL.
 - (iii). FORMAT.
 - (iv). REN.
9. A message 'Data Fault Error writing drive C:' is displayed on the screen while trying to save a file. Explain this fault and how you could solve it.
10.
 - (a). How do you list all files in the current directory with a Pause when the screen fills up?
 - (b). How do you list all files in the current directory using the Wide format?
 - (c). How do you list files in drive A:?
 - (d). How do you list all files starting with DB?
 - (e). How do you list all files ending with .EXE?
 - (f). What command sequence would you use to list as many files on the screen as possible?
11.
 - (a). How would you copy all the files which ends with .WKI and .DBF from the root directory in your hard disk to your floppy disk?
 - (b). How would you erase all data files in your disk which ends with .COM?
 - (c). How would you copy the file named TEST.PRG from the hard drive to your floppy disk under the following names?
 - (i). TEST1.PR
 - (ii). TEST1.GGR
 - (iii). TEST3.01
 - (d). Write a command sequence that will erase all the files in disk A starting with TEST regardless of their extension.
 - (e). Write any command sequence which will delete all files in drive A.
12.
 - (i). Give the DOS command used for formatting ordinary and system diskettes in drive A.
 - (ii). Why should precautions be taken while formatting diskettes?
13.
 - (a). Give the command that can be used to display the contents of a file named PROFIT.PRN from disk drive A:
 - (b). Give the command that can be used to display the contents of SALES.PRN on the screen.

BATCH FILES.

What are Batch Files?

Batch files are normal text files that contain a series of commands, which are executed whenever you type the name of the batch file.

Batch files must have the extension .BAT for DOS to recognize them as batch files.

Batch program - An ASCII (text) file that contains one or more MS-DOS commands. A batch program's filename has a .BAT extension. When you run a batch program, the commands are processed sequentially.

CONFIGURING THE SYSTEM.

Configuring the system means setting it up so that MS-DOS, your hardware and your applications work the way you want them to.

Most of your system's configuration information is stored in 2 files;

1. The **Autoexec.bat** file, and
2. The **Config.sys** file.

CONFIG.SYS File.

It is a text file containing commands that can configure (organize) your computer's hardware components (Memory, Keyboard, Mouse, Printer) so that DOS & your applications can use them.

The purpose of the CONFIG.SYS is to load installable device drivers & reserve space in the system memory for information processing. It instructs DOS on how you want it to work with the various components of your system configuration.

When MS-DOS starts, it carries out the commands in your CONFIG.SYS file first.

CONFIG.SYS File Commands.

Command	Purpose
BUFFERS	- Reserves storage areas equivalent to .5KB each in the Main memory area for transferring information to & from disks.
COUNTRY	- Allows you to change the national language setting for your system.
DEVICE	- Loads an installable Device driver (a program that controls a hardware component, such as a Mouse or CD-ROM drive) into Conventional memory.
DEVICEHIGH	- Loads an installable device driver into Reserved (Upper) memory.
❖ <i>To load DOS into Extended memory;</i>	
Device=C:\DOS\HIMEM.SYS ↵ (Himem.sys is the XMM memory manager).	
DOS=HIGH	
DEVICE=C:\DOS\EMM386.EXE 2048 - Simulates extended memory as Expanded memory.	
FASTOPEN.EXE	- Speeds up file access time by opening frequently used files & tracking their locations in memory.
DOS	- Specifies whether MS-DOS will use the <i>High Memory Area</i> (HMA) & whether it will provide access to <i>Upper Memory Blocks</i> (UMBs).
LASTDRIVE	- Sets the number of valid drive letters.
REM	- Indicates that the text that follows is disabled .
STACKS	- Specifies how much memory to reserve for processing hardware interrupts.

MODIFYING CONFIG.SYS File.

1. At A:\>, type:

Copy CON Config.Sys C:↵ (To create & copy the Config.sys file from drive A to C).

Break=ON {ENTER}

Buffers=20{ENTER}

Files=30{ENTER}, then press **F6**{ENTER}

With disk in drive A, press **CTRL+ALT+DEL** (To restart the computer).

2. At A:\>, type:

EDIT Config.Sys↵

Device=C:\DOS\Smartdrv.Sys↵ - Improves computer speed by putting information in a **Disk Cache** (the space in memory where the information from a disk is stored).

Install=C:\Fastopen.Exe C:=50↵ - Speeds up file access time by opening frequently used files & tracking their locations in memory rather than from the directory on the disk. 50 files will be traced.

Press **ALT+F**, and choose **EXIT**.

3. At A:\>, type:

Copy CON Config.Sys C:↵ (To create & copy the Config.sys file from drive A to C).

Buffers=30{ENTER}- Sets the no. of **Buffers** (Memory areas used for temporary holding file data as you read & write).

Files=40{ENTER}- Sets the no. of files that can be opened at one time.

Device=C:\DOS\Mouse.sys{ENTER}- Loads the mouse driver so that a mouse can be used with the system.

Press **F6**{ENTER}

With disk in drive A, press **CTRL+ALT+DEL**.

4. **Copy CON: C: Config.Sys**↵

Device=DOS\C:\Windows\Himem.sys↵

Device=Mouse\Mouse.com↵

Device=CDROM\CDROM DRV.SYS↵

Buffers=30↵

F6 {ENTER}

Boot {Press CTRL+ALT+DEL}

Examples of CONFIG.SYS Files.

1. **Device=C:\DOS\SETVER.EXE** - Loads the SETVER.EXE device drivers. The SETVER.EXE driver manages the MS-DOS version table.

Device=C:\DOS\HIMEM.SYS - Loads the HIMEM.SYS device drivers. The HIMEM.SYS driver manages Extended memory.

Device=C:\DOS\EMM386.EXE RAM - Loads the EMM386.EXE device drivers. The **RAM** switch directs the EMM386.EXE driver to provide access to Upper memory and simulate Expanded memory.

DeviceHigh=C:\Mouse\Mouse.Sys↵ - Loads the MOUSE.SYS device drivers, which provides access to the mouse into Upper memory.

Buffers=20 – Reserves 20 buffers for transferring information to & from disks.

Files=40 – Gives MS-DOS access to 40 files at one time.

Break=ON – Checks frequently for the CTRL-C or CTRL-BREAK key combination.

DOS=High,UMB – Runs MS-DOS in the High memory area and gives programs access to the Upper Memory Area.

2. **Files=99**

Buffers=40

Device=C:\DOS\Mouse.sys

Device=C:\DOS\Himem.sys

DOS=High

AUTOEXEC.BAT file.

It is a special batch program that runs every time the computer starts. It can be used to display the DOS Prompt in a particular format, specify the path, display screen messages and start programs (usually memory-resident programs such as Virus guards, device drivers, etc).

AUTOEXEC.BAT File Commands.

Command	Purpose
PROMPT	- Sets the appearance of the DOS Prompt.
MODE	- Sets the characteristics of your Keyboard, Monitor, Printer & Communication ports.
PATH	- Specifies the directories that MS-DOS should search for Executable files (files with .COM, .EXE & .BAT extensions) & the order in which the directories should be checked.
Echo Off	- Directs MS-DOS not to display the commands in the AUTOEXEC.BAT file as they run. You can also prevent a command from appearing by inserting an <i>at sign</i> (@) before the command.
SET	- Creates an environment variable that programs can use.

MODIFYING AUTOEXEC.BAT File.

1. At A:\>, type:

Copy CON Autoexec.Bat C: ↵ (To copy the Autoexec.Bat file from drive A to C).

Echo ON ↵

Date ↵

Time ↵

Path=C:;C:\123 ↵, then press **F6** ↵

With disk in drive A, press **CTRL+ALT+DEL** (To restart the computer).

2. At A:\>, type:

Copy CON Autoexec.Bat C: ↵

Echo OFF ↵

Path=C:;C:\Wp51;C:\Games ↵, then press **F6** ↵

With disk in drive A, press **CTRL+ALT+DEL**..

3. At A:\>, type:

Copy CON Autoexec.Bat C: {press ENTER}

Echo ON {press ENTER}

Date{press ENTER}

Time{press ENTER}

Path=C:;C:\123;C:\Dbase;C:\Wp51;C:\Graphics ↵, then press **F6**{press ENTER}

With disk in drive A, press **CTRL+ALT+DEL**.

Examples of AUTOEXEC.BAT Files.

1. **@ECHO OFF** - Tells DOS not to display the commands as they are being executed.
Path C:\DOS;C:\Windows - Sets the PATH to the DOS and WINDOWS directories.
Prompt=\$P\$G - Sets the DOS prompt to the familiar **C:\>** prompt.
CLS - Clears the screen.
C:\Wp51\Wp - Starts the WordPerfect program.

2. **DOS\SmartDrv. Exe**
DOS\DosKey.COM
Path=C:\;C:\123r24;C:\Dbase;C:\Wp51;C:\Word;C:\Graphics;C:\DOS;C:\QPro;C:\Windows

EDITING THE CONFIG.SYS & AUTOEXEC.BAT FILES.

Config.Sys & Autoexec.Bat files tell your computer what programs and devices to load on startup. E.g., a virus-scanner program to Autoexec.bat file may direct your computer to automatically load.

To remove or disable such a program or device, you may need to edit the Config.sys and/or Autoexec.bat files.

To edit your Config.Sys & Autoexec.Bat files;

1. In Windows 95 or 98, click **Start**, click **Run**, type **SYSEEDIT**, then press **ENTER**.
2. In the Config.sys or Autoexec.bat dialog box, type **REM** at the beginning of any line(s) that you want to disable.
3. Save changes & restart your computer.

NB: If you are working in MS-DOS;

1. Press **F8** at '*Starting MS-DOS...*', then choose **Command Prompt only**.
2. To edit the Autoexec.bat file, type:

EDIT Autoexec.bat ↵

3. Type **ALT-F-X** to exit, and then save the file when prompted.
4. Reboot your computer.

Descriptive Questions.

1. (a). What are batch files?
(b). What are the AUTOEXEC.BAT and CONFIG.SYS files used for?

Time Out - If a device is not performing a task; the amount of time the computer should wait before detecting it as an error.

Time Slice - The amount of processor time allocated to an application, usually measured in *Milliseconds*.

BASIC FUNCTIONS OF MS-DOS AS AN OPERATING SYSTEM.

1. It controls the functions of the computer.
2. It reports the status of the computer.
3. File manipulation:
It enables the user to delete files/directories & also to print files within the computer storage to a Printer.
4. It prepares new floppy disks, e.g., formatting.
5. It acts as an interpreter of all the commands.
6. It controls Input & Output devices.
7. It determines how data will be processed.
8. It provides Language translators.
9. It controls memory management.
10. It organizes information to facilitate easy storage & retrieval of documents. It allows the user to see & keep track of the number of files in the disk.

Exercise.

1. Describe the functions of the Disk operating system software in the computer system.

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